

QUANTILE REGRESSIONS AND VALUE AT RISK

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Limitations of Classical Regression Analysis

- Classical regression analysis is concerned with predicting the mean value of the dependent variable on the values of the independent variables
- In the presence of outliers and fat tails, the inferences made at the average may not apply to the entire distribution of returns
- Events such as global financial crises make modelling left-tail return distributions (unfavourable returns) all the more important
- A quantile regression involves estimating the functional relations between variables for all portions of the probability distribution
- One can establish the relationship between the dependent and independent variables for a particular quantile to manage the tail risk better

Linear Median Regression Model

The linear median regression model

$$Y = X'\beta + \epsilon$$

$$\text{median}[\epsilon|X] = 0$$

The model resembles the linear regression model but is different. The coefficient β in the median.

in the linear median regression model the median regression coefficient minimizes $M(\beta) = E|Y - X'\beta|$, the expected absolute error.

The sample estimator of this function is the average of absolute errors

$$M_n(\beta) = \frac{1}{n} \sum_{i=1}^n |Y_i - X_i'\beta|$$

Linear Median Regression Model

This is similar to the classical average of squared errors function but instead is the average of absolute errors.

β is the minimizer of $M_n(\beta)$:

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} M_n(\beta)$$

This is called the Least Absolute Deviations (LAD) estimator of β .

The LAD estimator $\hat{\beta}$ does not have a closed form solution so must be found by numerical minimization.

Quantile Regression

The mean and median are measures of the central tendency of a distribution. A measure of the spread of the distribution is its quantiles.

Recall that for $\tau \in [0, 1]$ the τ^{th} quantile q_τ of Y is defined as the value such that $P[Y \leq q_\tau] = \tau$.

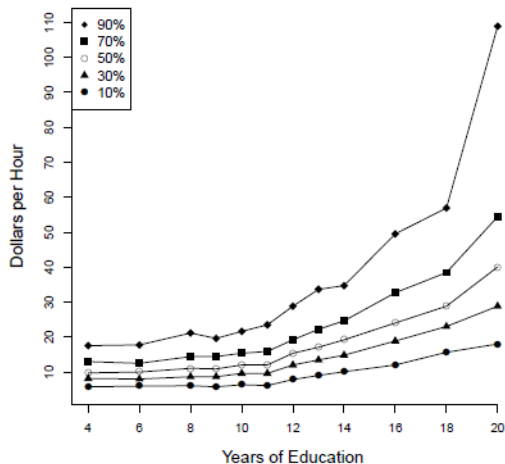
The conditional quantile of Y given $X = x$ is the value $q_\tau(x)$ such that $P[Y \leq q_\tau(x) | X = x] = \tau$.

The function $q_\tau(x)$ is also called the quantile regression function.

The conditional quantile function $q_\tau(x)$ is monotonically increasing in τ , thus if $\tau_1 < \tau_2$ then $q_{\tau_1}(x) < q_{\tau_2}(x)$ for all x .

Quantile Regression

Figure 1: Wage Quantile Regression



Source: Hansen (2021).

Quantile Regression

The linear quantile regression model is

$$Y = X'\beta_\tau + \epsilon$$

$$Q_\tau[\epsilon|X] = 0$$

Recall that the mean minimizes the squared error loss and the median minimizes the absolute error loss. There is an analog for the quantile. Define the tilted absolute loss function:

$$\rho_\tau(x) = \begin{cases} -x(1 - \tau) & \text{if } x < 0 \\ x\tau & \text{if } x \geq 0 \end{cases}$$

For $\tau = 0.5$ this is the scaled absolute loss $\frac{1}{2}|x|$. For $\tau < 0.5$ the function is tilted to the right. For $\tau > 0.5$ it is tilted to the left.

The function $\rho_\tau(x)$ has come to be known as the **check function**.

Quantile Regression

The coefficient β_τ minimizes $M(\beta, \tau) = E[(\rho_\tau(Y - X'\beta))]$, the expected check function loss.

The estimator of this function is the sample average

$$\hat{\beta}_\tau = \underset{\beta}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^n \rho_\tau(Y_i - X_i'\beta)$$

This is called the Quantile Regression estimator of β_τ .

The coefficient $\hat{\beta}_\tau$ does not have a closed form solution so must be found by numerical minimization.

Value at Risk (VaR)

- A measure of (quantifiable) risk
- Elements associated with a VaR
 - ▶ A PDF of profits and losses of a company, or the value of a portfolio, etc.
 - ▶ A horizon
 - ▶ A critical value (typically 95%)
- The VaR is the maximum loss with 95% confidence
- $VaR_{0.95} = X$ such that $Pr\{X \leq x_{0.05}\} = F(x_{0.05}) = 0.05$, where F is the cumulative probability function.

Value at Risk (VaR)

Figure 2: Value at Risk (VaR)



Note: The actual value of the VaR depends critically on the function used.

Value at Risk (VaR)

VaR: Estimation

- Parametric (analytical) VaR
- Historical VaR
- Empirical VaR
- Monte Carlo VaR
- Non-Parametric VaR

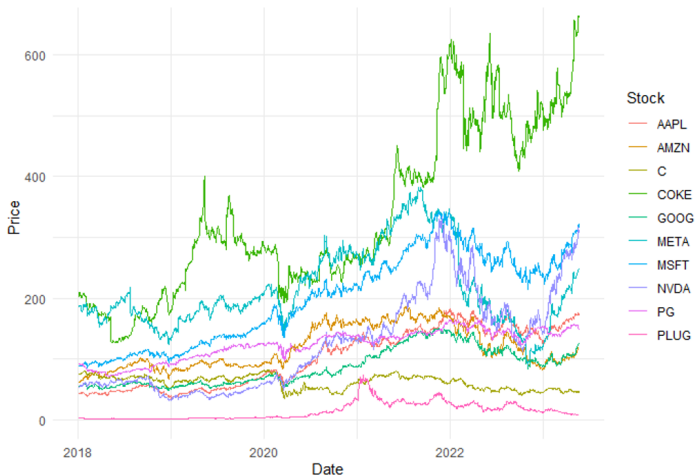
Parametric (analytical) VaR

- $P_1, P_2, \dots, P_T$ monthly profits (or losses)
- Assume that they follow an iid $N(\mu, \sigma^2)$. Estimate $\hat{\mu}, \hat{\sigma}^2$
- Thus, VaR is x such that $\Phi = \left(\frac{x_{0.05} - \hat{\mu}}{\hat{\sigma}} \right)$
- The estimate is model-dependent. In the case of a normal distribution, it assumes thin tails

Parametric (analytical) VaR

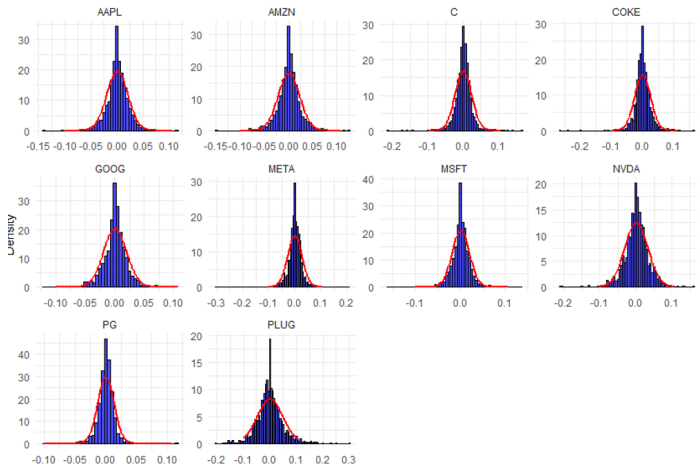
Suppose we have a portfolio of USD 1 million which is invested in 10 stocks weighted equally.

Figure 3: Time series plot of stock prices



Parametric (analytical) VaR

Figure 4: Histogram of daily returns of stock prices and their normal density



Source: Bloomberg.

Parametric (analytical) VaR

Trade-off in the size of the time window of the sample.

A larger window allows to have more observations and capture the underlying distribution of returns more accurately (important to capture extreme events).

The assumption that the distribution of returns does not change during the sample period difficult to hold.

The value at risk of a single stock is given by the 1% quantile of the estimated (normal) distribution. However, for the whole portfolio the covariance structure must be considered.

The mean of the portfolio is $w'\mu$ and the variance of the portfolio is $w'\Sigma w$.

Parametric (analytical) VaR

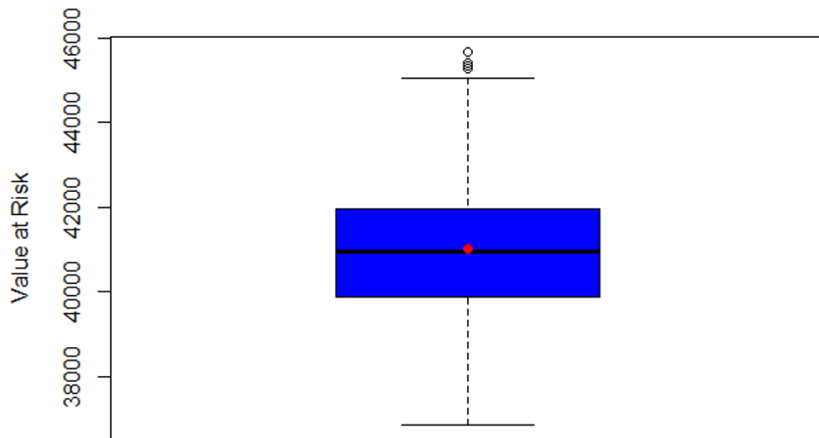
A portfolio of 1 million that gives equal weight to the 10 stocks has a 1-day portfolio variance of 1.76%, which means that the 1% value at risk in terms of returns is a decrease of 4.10% or in terms of the portfolio, the 1% Value at risk (VaR) is \$41,019.94

An easy way to calculate the error around this estimation is with the bootstrapping, which means sampling from the sample and generating VaR using the same method

The next plot shows the boxplot (quartiles) of the VaR calculated from 1000 bootstrap iterations and the **Variance Covariance method**.

Parametric (analytical) VaR

Figure 5: Bootstrapped boxplot - Variance-Covariance method



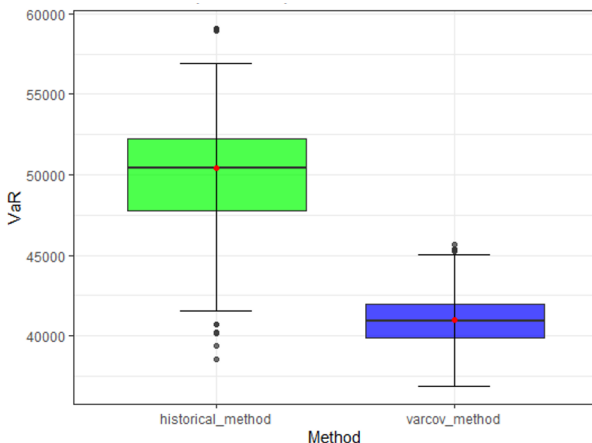
Historical VaR

- $P_1, P_2, \dots, P_T$ monthly profits (or losses)
- Order $P_{(1)}, P_{(2)}, \dots, P_{(T)}$ from smallest to largest
- Choose the highest of the lowest 5% losses
- It does not assume a specific distribution
- It assumes that the historical time series is quite informative about the future profits and losses

Historical VaR

The 1% VaR gives a negative return of 5.04% or in terms of the portfolio the 1% VaR is \$50,420.38. We can again resample from the historical sample using bootstrap and generate a boxplot.

Figure 6: Bootstrapped boxplot - historical method



Empirical VaR

- $P_1, P_2, \dots, P_T$ monthly profits (or losses)
- Estimate empirical $\hat{F}(p) = \frac{1}{T} \sum_{t=1}^T 1\{P_t < p\}$
- VaR is x such that $\hat{F}(p_{0.05})$
- This approach has the advantage that it does not assume a specific distribution

Monte Carlo VaR

- $P_1, P_2, \dots, P_T$ monthly profits (or losses)
- For the analytical VaR, we assumed iid distribution
- We could use an AR model: $P_{t+1} = \rho_0 + \rho_1 P_t + \epsilon_{t+1}$
- We use the sample to estimate AR
- Simulate ϵ_{t+1} with Monte Carlo to back out P_{t+1} (different to Bootstrap)
- The basis of Monte Carlo simulation is the simulation of a uniform distribution

Monte Carlo VaR

When we have a portfolio, we can follow the next steps:

- We model the marginal distribution of each variable. For example, let's model the returns as t -distributions
- We transform the variables to uniform distributions using the inverse of their distribution function. We now have a collection of $n = 10$ uniform variables
- We select a copula that models the dependence structure of our variables
 - ▶ A copula is a multivariate cumulative distribution function for which the marginal probability distribution of each variable is uniform
 - ▶ In the context of financial crises, it is common for asset returns to exhibit "tail dependence". This is the phenomenon where extreme events in one asset class (e.g., a large loss) are often associated with extreme events in another asset class

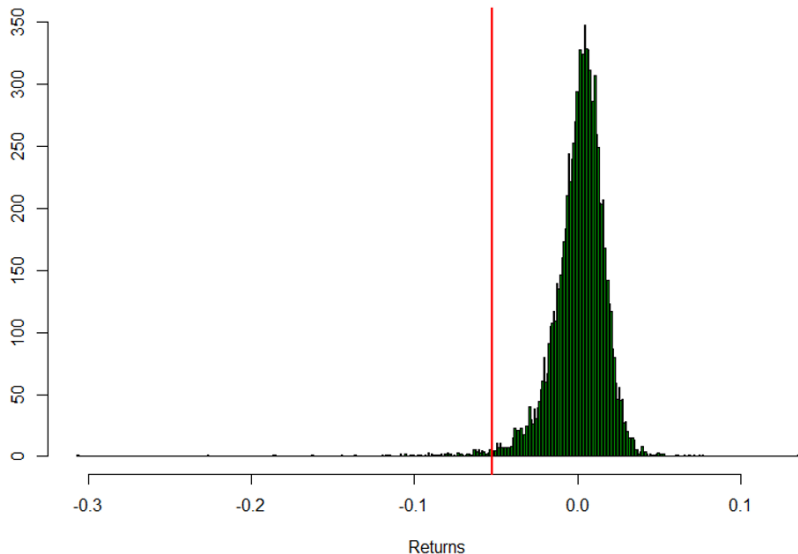
- In the context of financial crises, it is common for asset returns to exhibit "tail dependence". This is the phenomenon where extreme events in one asset class (e.g., a large loss) are often associated with extreme events in another asset class
- Clayton copula is "left tail dependent"; that is, it implies strong correlation in the left tail and low correlation in the right tail of the distributions
- Clayton copula has a single parameter (θ) which gives the strength of the dependence between the random uniform variables

Monte Carlo VaR

- The fitted copula in the example is $\theta = 0.5018$
- With the fitted copula simulate 10,000 observations of uniform data
- Transform the uniform data back to t -distributions using the distribution function for each variable
- Weight each simulated return in the portfolio
- Calculate the 1% VaR with the simulated returns
- The result is that the 1% value at risk in terms of returns is a decrease of 5.24%. In terms of the portfolio, the 1% Value at risk is \$52,449.86

Monte Carlo VaR

Figure 7: Histogram of returns - Clayton copula



Non-Parametric (Kernel) VaR

- $P_1, P_2, \dots, P_T$ monthly profits (or losses)
- Estimate a Kernel distribution $\hat{f}(P) = \frac{1}{Th} \sum_{t=1}^T \mathcal{K}\left(\frac{P-P_t}{h}\right)$
- We discussed that there are two elements to the kernel density, the function of the kernel itself and the bandwidth
- Thus, VaR is x such that $\hat{F}(x_{0.05})$

Quantile Regression and VaR

Recall the linear quantile regression model

$$R = X'\beta_\tau + \epsilon$$

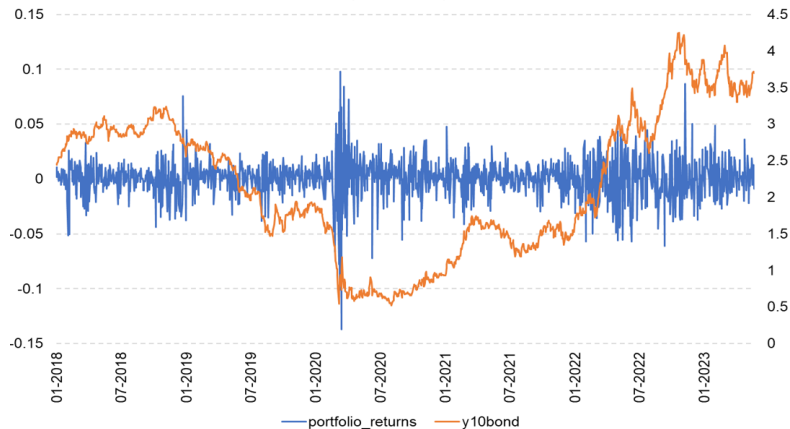
$$Q_\tau[\epsilon|X] = 0$$

where R is the returns (portfolio returns), X is the matrix of explanatory variables (bond yields, inflation, GDP growth).

For instance, we would like to see the effect of these exogenous variables on the distribution of portfolio returns.

Quantile Regression and VaR

Figure 8: Portfolio returns and 10-year bond yields



Quantile Regression and VaR

Table 1: Dependent variable: Returns

Variable	Coefficient	Std. Error	p-value
Intercept	-5.4181	1.269	0.00002
Inflation	-0.1141	0.06494	0.07898
GDP growth	0.19073	0.16342	0.24337
10-year bond yields	0.63175	0.45480	0.16503

- An increase of inflation of 100bp reduces the expected return at the left tail by 11bp
- For a portfolio of 1 million, this means that the 1% VaR increases by \$1,141 for every increase in inflation of 100bp
- In other words, higher inflation increases the risk at the left tail of the distribution

Extensions

The flexibility of this methodology has lead to research on applications of the VaR methodology in economics and finance:

- Growth at Risk, Capital Flows at Risk
- Expected shortfall. An alternative to value at risk that is more sensitive to the shape of the tail of the loss distribution. also called conditional value at risk (CVaR)
- CoVaR. Useful to study systemic risk (how to measure the potential contribution of individual banks)
- Δ CoVaR. Useful to study the change of the system's VaR given an institution condition of distress
- Entropic value at risk (EVaR)
- Range Value at Risk (RVaR)

References

- CEMLA (2022, 2021). *Course on Financial Econometrics*.
- Hansen, B.E. (2022b). *Econometrics*. Princeton University Press.
- Linton, O. (2019). *Financial Econometrics Models and Methods*. Cambridge University Press.